

A Pre-feasibility Study for the Establishment of a Zeolite Manufacturing Plant



2017



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1. Executive summary

Zeolite ore is characterized by physical properties and chemical composition, which allows it to be used in various industrial fields such as fertilizer industry and in the field of water purification, as well as, in the medical field, which is used to purify ammonia in the dialysis unit, so there is an opportunity to invest in exploiting this ore by extraction and treatment using physical methods such as magnetic separation and hydrocyclone.

Al-Mafraq Governorate was chosen because it is considered one of the governorates where natural resources are available for attracting projects and activities. The existing projects and activities are: (19) quarries of building stone, (6) rubble quarries concentrated in Al-Ruwaished district and (5) (Al-Buzlana) and Zeolite exploitation companies in the Safawi and Al Mafraq district area, since the governorate had many discovered mineral materials such as building stone, volcanic tuff and zeolite, where the reserves amounted to about 470 million tons, of which limited quantities of 15 thousand tons are exploited per year for the production of cement, limestone and white cement.

Table1 : Project Main Indicators

Item	JD
Total Investment cost	1,836,118
Internal of Return Rate	31.54%
Payback Period (Years)	3
Net Present Value	3,155,719
Job Oppurtunities	10
Projected Annual Revenue for 10 years	39,550,384
Total Operating Cost (10 years)	31,976,485
Gross Profit (10 years)	3,990,802

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

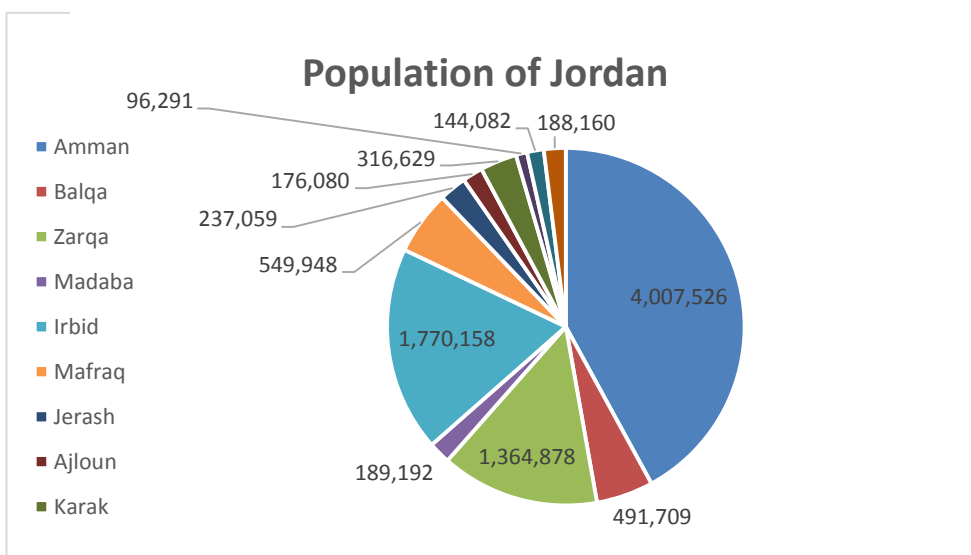
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1: Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

Department of Statistics (DOS), 2015.

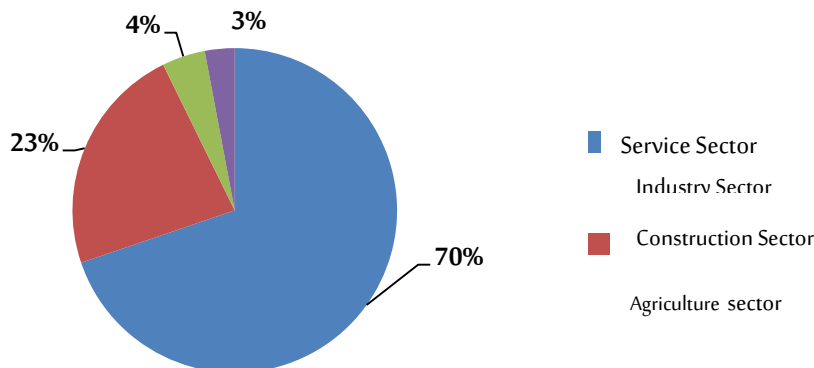
Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to

encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in

licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special

economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project description

The Zeolite ore has a wide spread in the Jordanian areas with a good economic rate and distinctive qualities. It is located in the Syrian area of Harat Al Sham, in addition to volcanic tuff. Therefore, the idea of mining and processing of this ore through mechanical separation methods to increase the percentage of zeolite, which is estimated to reach 90%.

Zeolite

Represents a series of alkaline elements and hydro aluminum silicate, these minerals are most commonly found in sedimentary rocks of volcanic origin with high porosity; small granular size and solubility. Zeolite is characterized by its ability to replace ions without any change in its structural structure and its ability to replace ions and absorb moisture to be used in environmental, agricultural and industrial aspects. The first zeolite was discovered in Jordan in 1987. The Jordanian zeolite is produced by the change of volcanic glass from the Artine volcano in northeast Jordan, which covers the free basalt plateau of Harat Al Sham about 15,000 km² and extends northwest / southeast of the southern part of the Damascus floor and southward to Northeast of Jordan and to the northwestern part of Saudi Arabia

Chemical composition

The Jordanian zeolite is composed of Phylbsite, Fuzsite, and Shabazite minerals, which are formed by the interaction of basalt glass with alkaline water. Jordanian zeolite is estimated at volcanic tuff between 20-65%. The following table shows the most important chemical structures of Jordanian zeolite

Table 3: Chemical composition of zeolite ore

Area / Chemical composition %	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	MgO	CaO	K ₂ O	Na ₂ O
Tall rmah	42.0	12.8	12.1	10.1	8.5	0.8	4.0
Mkawer	42.7	13.9	12.7	9.2	9.8	1.9	2.1
Alarten	38.6	12.8	12.1	9.6	9.3	1.5	2.1
Shehan	44.0	13.2	8.3	8.6	11.3	1.2	2.1
Tall chohera	35.0	10.2	11.3	7.6	20.2	0.7	2.4
Jabal alatata	48.0	10.8	8.1	7.7	10.7	0.5	1.5
Tlol alshhbaa	41.7	11.8	12.0	10.3	9.4	1.7	2.8
Jabal enaza	40.0	70.9	8.8	8.6	15.8	0.9	5.7

Source: Natural Resources Authority

Project Objectives

The objectives of the project are summarized as follows:

- Utilization of zeolite ore exists in the Jordanian areas since it can be used in many industries and after processing.
- Providing employment opportunities for the local community.
- Increasing competitiveness in the local markets and upgrading the quality of raw materials.
- Increasing the export volume.

Proposed service

Preliminary studies indicated that there is a large reserve, the existing extraction and manufacturing is used only in the field of agriculture, based on the chemical composition and features of zeolite, it can be used in many areas and uses such as:

- Purification of water and wastewater from ammonia in residential areas and cities
- To be added to organic fertilizers to increase their effectiveness
- It is used as a dryer as a result of its ability to absorb water to be used as additive to chemical fertilizer and feed to avoid forming hardening blocks during storage and prevents rotting in the poultry feed.
- Exploitation in Solar energy for heating and cooling the water
- Used as a filler in the paper industry
- Used to clean ammonia in dialysis unit
- Used to remove strontium and cesium from nuclear waste
- Used in manufacturing of pure oxygen

Target Segments

There are many industries use zeolite as one of the raw materials in its products such as fertilizer industry, medical aspects, and water purification plants.

Fertilizer Companies:

- United Fertilizer Company
- Al Rowad Fertilizers Company
- Jordan Phosphate Company
- Others

Water purification plants:

- Al Samra Station
- Zi Station
- Shoaib Valley Station
- Others

Pharmaceutical companies

- Dar Al-Dawa Company
- Alhekma Pharmaceuticals Company

- Atlas Medical Factory
- Others

Site analysis

The northern Arabian basalt plateau covers an area of 11,000 km² of the land of northeast Jordan. The following map shows the location of the Jordanian zeolite ore.

Figure 3: The locations of zeolite in Jordan



Market analysis

Investments in the field of mining and processing of zeolite are open to all by due to huge quantity of zeolite reserves in Jordan. The following table shows the amount of reserves of Jordanian zeolite ore.

Table4 : Reserves of Jordanian zeolite ore

Area	Reserve(million/ton)
Tall remah	46.0
Alarten	170.0
Tlol alshhba	9.2
North-East Region	472.0
Other	134.0
Total	831.2

Source: Natural Resources Authority

Competitor analysis

The annual production of the Jordanian zeolite is estimated at 2,626,128 tons. There are many companies producing the raw material such as:

Table5 : Competitors Companies

Company/Year	2012	2014	2016
Emaar Mining Company	The company acquired mining right in 2014	34194	42986
Green Technology Group	14844	19658	29685

Galilee Mining Company	8743.34	23897.77	52973
Eid Al-Mughira Company	5486	7969	11698
Technical Company for Mining Investments	20947	38152	63142
International Company for Mining and Agricultural Materials	36883	180004	276302

Price analysis and pricing policy

The price of natural zeolite ton is between 35-70 JD and the price reaches JD 425 for industrial zeolite. The zeolite production started in Jordan in 1998, therefore it is considered to be one of the modern mining sectors. The total production in 2002 was 4,500 tons, the local consumption of zeolite is about 3500 tons and the balance is exported abroad. The export volume for 2014 was estimated at 19,297 tons. The annual production of Jordanian zeolite mining and processing companies amounted to 2,626,128 tons / year for the year 2016

Table 6: Price Analysis

Type	Ratio (%)	Partial production (tons)	Price (JD)	Revenues JD
Zeolite	90	67,500	50	3,375,000
Volcanic taf	10	7500	10	75,000
Total	100	75,000	-----	3,450,000

The Jordanian zeolite can be sold at a price of up to JD 425 through advanced processing to get industrial products such as Calcite, Phylpsite, Gypsum and Phosphate.

Market share Analysis

Based on the production capacity of the mining companies which are specialized in the mining and processing of zeolitewhich amounts as 2,626,128 tons per annum, the production capacity of the proposed plant is estimated to be 75,000 tons per year, 250 tons per day assuming 300 working days per year and a market share of 2.8%.

Projected Revenues Analysis

- The expected revenues for 10 years are as follows:

Table 7: Revenues of the project during the years 2018-2027

Item	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Selling price of zeolite	50	52	53	55	56	58	60	61	63	65
Selling price of volcanic tuff	10	10	11	11	11	12	12	12	13	13
Zeolite Revenues	3375000	3476250	3580538	3687954	3798592	3912550	4029927	4150824	4275349	4403609
Volcanic Tuff revenues	75000	77250	79568	81955	84413	86946	89554	92241	95008	97858
Total Revenues	3450000	3553500	3660105	3769908	3883005	3999496	4119480	4243065	4370357	4501467

The projected annual revenues for the project were calculated based on the analysis of the production capacity of the project as the following table shows:

Table8 : Production Capacity

The Analysis of the Production Capacity	
The size of the raw material extracted	75000
Percentage of extracted zeolite	90%
Percentage of volcanic tuff extracted	10%

4. Technical analysis

Required Human Resources

Table 9: Man Power required for project

Job Title	Number
Indirect employees	
Project manager	1
Accountant	1
Security guard	2
Technician	1
Driver	3
labor	2
Total	10

The following is an explanation of the labor force:

Project Manager:

- Works independently and within the authority given to him and proposes policies, work plans, representing the company, planning and management of the work in the factory.
- Follow up the implementation of plans and programs in accordance with the policies and goals of the company to optimize the interests of owners.
- Supervising the administrative and financial affairs in accordance with the established instructions in this regard.
- Managing subordinates and developing their skills.
- Develop procedures, ensure safety, occupational health and public facilities

The accountant:

- Recording all financial movements of a project
- Preparation of financial statements
- Preparation and calculation of salaries and all related to tax and social security
- Any other tasks determined by the direct supervisor in proportion to the nature and work aspects.

The mining technician:

- Work on the operation of production lines and follow-up work.
- Implementation of daily production plans, to ensure the implementation of the quantity and quality required.
- Report any damage or malfunction in the machine or production.

The driver:

- Driving vehicles to carry out the tasks assigned to him and applying road safety instructions.
- Record the tasks he performs and the distances he commits in the private register
- Follow up the maintenance of vehicles of the project.

The guard:

- Implementation of the security instructions adopted to maintain the security system in the project.
- Ensure that all departments are closed at the end of the work and that all employees leave the site or sections.
- Touring sites during the guarding period.
- Ensure that there is an authorization to transfer any materials or goods from the site

Technical requirement analysis

Land and construction Works

The plant will be located in Al Mafraq, located in the north area of the Kingdom, which is strategically located and the land will be leased from concerned authority in the governorate.

Table 10: Land required for the project

Land	Area m ²
Land required	50,000

The required construction work for the project is as follows:

Table 11: Required construction works

construction works	Area m ²
Construction of Hangar	1,000
Building Works	150

Machinery and Equipment

The following table shows the machinery and equipment required for the project as follows:

Table 12: The Required Machinery and Equipment

Machine	Number
Loder	1
Crushing machine	2
Magnatic sepration machine	1
Seiving machine	1
Dumber	1

Furnitures

The project requires integrated office furniture, such as administrative offices, air conditioners and others.

Vehicles and Transportation

The project needs the following vehicles and transportation:

Table 13: The Required Vehicles

Vehicles and transportation	Number
Pick up	1

Description of production process and steps of manufacturing

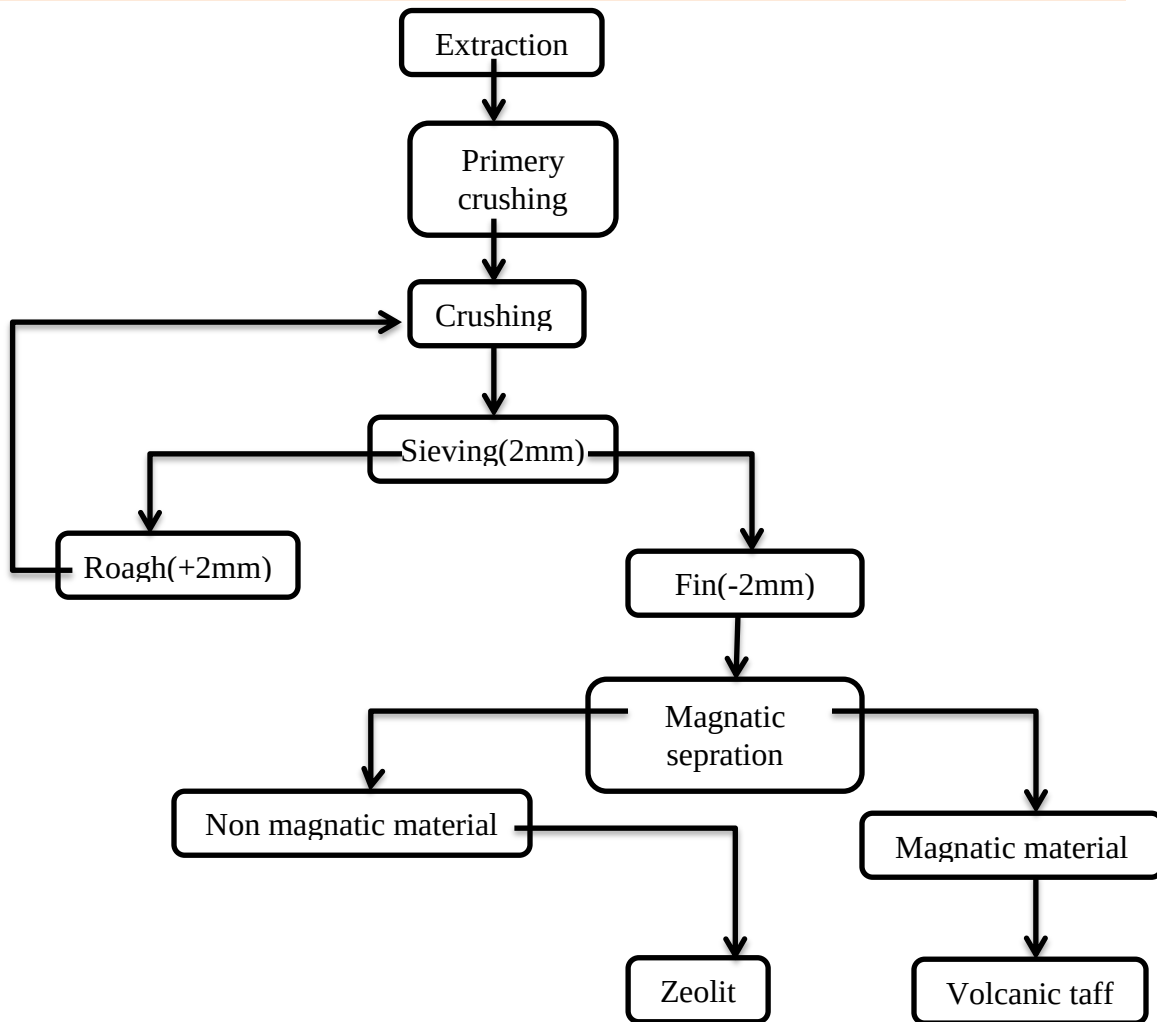


Figure4 : Description of the production process

- **Crushing:** After the extraction of ore from its sites is introduced to the initial cracking process by jaw crusher to conduct cracking operations for this ore to obtain a granular size 5 mm, followed by secondary crushing operations to obtain a granular size of 2 mm.
- **Sieving:** This process precedes the hydrocyclion separation process so the secondary cracking products are separated into two parts, one less than 2 mm and the other greater than 2 mm which will return to the secondary crushing process
- **Hydrocyclonic Separation:** This process is used to get rid of the soft volumes in which clay minerals and some ferrous metals are concentrated, which hinder work in later stages
- **Magnetic separation:** This process depends on the magnetic force, which in turn separates the materials into two parts; magnetic materials and non-magnetic materials

5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 10.75% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 14: General Assumption

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 15: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 16: Annual Growth Rates Assumptions

	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 17: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	65.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	2.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table 18: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
Natural wealth extraction (JD / ton)	0.50
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 19: Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	20%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 20: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	734,447
Loans Value	1,101,671

Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	1,101,671
Loans Percentage	%60
Equity	
Cost of Equity	%15.5
Equity Value	734,447
Equity Percentage	%40
Gross	
Project Value	1,836,118
Weighted Average Cost of Capital	%100

Capital Expenditures

The estimated capital costs of the project are 1,836,118 JD and cover the cost of land, construction, machinery and equipment, furniture and furnishings, vehicles and transportation, pre-operating expenses and initial working capital. These costs are shown in the following tables:

Table 21: Construction Work Capital Expenditure

Construction Works	Cost (JOD per Sqm)	Area (sqm)	Gross Cost
Hanger	100	1,000	100,000
Construction Works	200	150	30,000
Other			13,000
Gross Cost			143,000

Table 22: Machinery and equipment Capital Expenditure

Machinery and equipment	Quantity	Unit cost (JD)	Gross Cost (JD)
Loder	1	60,000	60,000
Crushing machine	2	40,000	80,000
Magnatic sepration machine	1	400,000	400,000
Seiving machine	1	25,000	25,000
Dumber	1	200,000	200,000
Gross Total (JD)			765,000

Table 23 : Furnitutre Capital Expenditure

Furnitutre	Quantity	Unit cost	Gross Cost /Jordan Dinar
Office Furnitutre	-	-	13,000
Total			13,000

Table 24: Pre-operating expenses

Pre-operating expenses	Gross Cost
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 25: Initial working capital

Initial working capital	Number of months	Percentage	Cost (JD)
Operating expenses	3	%25	697,331
Administrative and general expenses	3	%25	4,850
Marketing expenses	3	%25	2,156
Indirect salaries	3	%25	18,360
Gross Total			722,698

Table 26: Summary of expenses

Summary of capital expenditures	Cost (JD)	Contingency	Cost (JD)	Percentage
Land	0	10%	0	%0.0
Construction Works	143,000	10%	157,300	%8.6
Machinery and Equipment	765,000	10%	841,500	%45.8
Furniture	13,000	10%	14,300	%0.8
Vehicles and Transportation	23,000	10%	25,300	%1.4
Pre-operating expenses	2,500	10%	2,750	%0.1
Initial working capital	722,698	10%	794,968	%43.3
Total (JD)	1,669,198		1,836,118	100%

Sources of funding

The financing structure required for the project consists of loans and equity. The investment cost of this project was estimated at JD 1,836,118. This amount is determined through fixed assets, which include land, construction, machinery and equipment, furniture and furnishings, vehicles and transport vehicles, pre-operating expenses and initial working capital. (60%) of the project will be financed through loans (1,101,671 JD) and the rest (40%) through self financing (equity). The following table shows the general structure of the required financing:

Table 27: Sources and use of finance

Sources of funding	amount	Percentage
Equity (self-financing)	734,447	%40
Loans	1,101,671	%60
Gross Total	1,836,118	%100
Use of funding	amount	Percentage
Capital expenditure	1,038,400	%56.55
Pre-operating expenses	2,750	%0.15
Working capital	794,968	%43.30
Gross Total	1,836,118	%100.00

Operating Expenses

The following table shows the expected operating expenses of the project:

Table 28: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw material	2,242,500	2,309,775	2,379,068	2,450,440	2,523,954	2,599,672	2,677,662	2,757,992	2,840,732	2,925,954
Service expenses	172,500	177,675	183,005	188,495	194,150	199,975	205,974	212,153	218,518	225,073
Maintenance	86,250	88,838	91,503	94,248	97,075	99,987	102,987	106,077	109,259	112,537
Consumables	34,500	35,535	36,601	37,699	38,830	39,995	41,195	42,431	43,704	45,015
Direct salaries	0	0	0	0	0	0	0	0	0	0
Other expenses	253,575	261,182	269,018	277,088	285,401	293,963	302,782	311,865	321,221	330,858
Gross Total	2,789,325	2,873,005	2,959,195	3,047,971	3,139,410	3,233,592	3,330,600	3,430,518	3,533,433	3,639,436

General and Administrative Expenses

The following table shows the expected administrative and general expenses of the project:

Table 29: administrative and general expenses

description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Communications expenses	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality expenses	600	618	637	656	675	696	716	738	760	783
Stationery and prints	600	618	637	656	675	696	716	738	760	783
Transportation and movements	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent expenses	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training expenses	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance expenses	7,788	8,022	8,262	8,510	8,765	9,028	9,299	9,578	9,866	10,162
Other expenses	1,764	1,817	1,871	1,927	1,985	2,045	2,106	2,169	2,234	2,301
Gross Total	19,402	19,984	20,583	21,201	21,837	22,492	23,167	23,862	24,578	25,315

Marketing Expenses

Table 30: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	8,625	8,884	9,150	9,425	9,708	9,999	10,299	10,608	10,926	11,254

Human Resources

The salaries and monthly wages of each employee were estimated to be multiplied by 16 months of the year for the purpose of calculating the increase in salaries and annual wages due to social security, health insurance, etc. to the employees of the administration. The following table shows the expected annual salaries and wages. The annual rate of increase in wages and salaries is taken into account:

Table 31: Expected monthly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Accountant	400	416	433	451	470	489	509	530	552	574
Security guard	300	312	325	338	352	367	382	397	414	431
Technical mining	350	364	379	395	411	428	445	464	483	502
Driver	380	396	412	429	446	465	484	503	524	546
labor	300	312	325	338	352	367	382	397	414	431

Table 32: Expected yearly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Accountant	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Security guard	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Technical mining	5,600	5,830	6,069	6,317	6,576	6,846	7,127	7,419	7,723	8,040
Driver	18,240	18,988	19,766	20,577	21,420	22,299	23,213	24,165	25,155	26,187
labor	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Total indirect salaries	73,440	76,451	79,586	82,849	86,245	89,781	93,462	97,294	101,283	105,436

Depreciations

The following tables show all equipment costs, expenses and depreciation rates

Table 33: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	0	%0.0	%0.0
Construction Works	157,300	%5.0	%0.0
Machinaries	841,500	%10.0	%2.0
Furniture	14,300	%10.0	%5.0
Vehicles	25,300	%10.0	%0.0
	1,038,400		

Table 34: Depreciation and annual increases on construction work

Summary of capital costs, additions and annual	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total fixed assets	1,038,400	1,055,945	1,073,862	1,092,161	1,110,848	1,129,9	1,149,4	1,169,3	1,189,6	1,210,4
Total consumption expenses	95,975	97,730	99,521	101,351	103,220	105,128	107,078	109,069	111,103	113,180
Total accumulated	95,975	193,705	293,226	394,577	497,797	602,925	710,003	819,072	930,175	1,043,3
Total additions	0	17,545	17,917	18,298	18,688	19,086	19,494	19,911	20,338	20,775
Total net book value	942,425	862,241	780,637	697,584	613,052	527,010	439,426	350,268	259,504	167,099
Loans										

The following table represents the annual loan amount and the annual payment method with the details of the remaining amounts of loans for each year.

Table 35: Loan Details

Loan Value	1,101,671
Annual Interest Rate	9.48%
Loans Period	5
Grace Period	0
Loan Starts at year:	2018
Annual Payment	286,767
Number of Payments	5

Year	Annual Payment	Interest	Equity capital	Loan Remaining Value
2018				1,101,671
2019	286,767	104,438	182,329	919,342
2020	286,767	87,154	199,614	719,728
2021	286,767	68,230	218,537	501,190
2022	286,767	47,513	239,255	261,936
2023	286,767	24,832	261,936	0

Income Statement

Table 36: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	3,450,000	3,553,500	3,660,105	3,769,908	3,883,005	3,999,496	4,119,480	4,243,065	4,370,357	4,501,467
Gross Operating Revenues	3,450,000	3,553,500	3,660,105	3,769,908	3,883,005	3,999,496	4,119,480	4,243,065	4,370,357	4,501,467
Operating Expenses	2,789,325	2,873,005	2,959,195	3,047,971	3,139,410	3,233,592	3,330,600	3,430,518	3,533,433	-3,639,436
	-	-	-	-	-	-	-	-	-	
Gross Operating Profit	660,675	680,495	700,910	721,937	743,596	765,903	788,881	812,547	836,923	862,031
Gross Profit Percentage	%19	%19	%19	%19	%19	%19	%19	%19	%19	%19
Salaries and Benefits (Indirect Staff)	-73,440	-76,451	-79,586	-82,849	-86,245	-89,781	-93,462	-97,294	-101,283	-105,436
General and Administrative Expenses	-19,402	-19,984	-20,583	-21,201	-21,837	-22,492	-23,167	-23,862	-24,578	-25,315
Marketing Expenses	-8,625	-8,884	-9,150	-9,425	-9,708	-9,999	-10,299	-10,608	-10,926	-11,254
Pre Operating Expenses	-2,750									
Gross Indirect Expenses	-104,217	-105,319	-109,319	-113,474	-117,790	-122,272	-126,928	-131,764	-136,787	-142,005
Income before Interest, Depreciation and Tax	556,458	575,177	591,591	608,463	625,806	643,631	661,953	680,783	700,136	720,026
Fixed Assets Depreciations	-95,975	-97,730	-99,521	-101,351	-103,220	-105,128	-107,078	-109,069	-111,103	-113,180
Income before Tax and	460,483	477,447	492,070	507,112	522,586	538,503	554,875	571,714	589,033	606,846

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Interests										
Bank Interests	-104,438	-87,154	-68,230	-47,513	-24,832	-0	-0	-0	-0	-0
Income Before Tax	356,045	390,294	423,840	459,599	497,754	538,503	554,875	571,714	589,033	606,846
Income Tax	-71,209	-78,059	-84,768	-91,920	-99,551	-107,701	-110,975	-114,343	-117,807	-121,369
Net Profit	284,836	312,235	339,072	367,679	398,204	430,802	443,900	457,371	471,227	485,477
Net Profit Percentage	%8	%9	%9	%10	%10	%11	%11	%11	%11	%11
Compulsory Reserves	-28,484	-31,223	-33,907	-36,768	-39,820	-43,080	-44,390	-45,737	-47,123	-48,548
Retained Earnings	256,352	537,364	842,528	1,173,440	1,531,823	1,919,545	2,319,055	2,730,689	3,154,793	3,591,722

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 37: Expected Cashflows Statement

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	284,836	312,235	339,072	367,679	398,204	430,802	443,900	457,371	471,227	485,477
Non-monetary adjustments										
Bank Interests	104,438	87,154	68,230	47,513	24,832	0	0	0	0	0
Depreciation	95,975	97,730	99,521	101,351	103,220	105,128	107,078	109,069	111,103	113,180
Total Operating Cashflows before Additions to Working Capital	485,249	497,118	506,823	516,543	526,255	535,931	550,978	566,440	582,330	598,657
Inventory (Increase/Decrease)	-186,875	-5,606	-5,774	-5,948	-6,126	-6,310	-6,499	-6,694	-6,895	-7,102
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	91,138	2,734	2,816	2,901	2,988	3,077	3,170	3,265	3,363	3,464
Working Capital (Increase/Decrease)	-95,738	-2,872	-2,958	-3,047	-3,138	-3,233	-3,330	-3,429	-3,532	-3,638
Net Cashflows from Operating Activities	389,512	494,246	503,865	513,496	523,116	532,698	547,648	563,011	578,797	595,019

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cashflows from Investments										
Activities										
Fixed Assets (Procurement)	-1,038,400	-17,545	-17,917	-18,298	-18,688	-19,086	-19,494	-19,911	-20,338	-20,775
Net Cashflows from Investments	-1,038,400	-17,545	-17,917	-18,298	-18,688	-19,086	-19,494	-19,911	-20,338	-20,775
Activities										
Cashflows from Financing										
Activities										
Capital	734,447									
Loan Amortization	-182,329	-199,614	-218,537	-239,255	-261,936	0	0	0	0	0
Bank Interest Rate	-104,438	-87,154	-68,230	-47,513	-24,832	-0	-0	-0	-0	-0
Loans	1,101,671	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing	1,549,350	-286,767	-286,767	-286,767	-286,767	0	0	0	0	0
Activities										
Net increase (decrease) in cash	900,462	189,933	199,180	208,431	217,661	513,612	528,154	543,099	558,459	574,243
Cashflows at the Beginning of Period	0	900,462	1,090,395	1,289,575	1,498,006	1,715,667	2,229,279	2,757,433	3,300,532	3,858,991
Cashflows at the end of Period	900,462	1,090,395	1,289,575	1,498,006	1,715,667	2,229,279	2,757,433	3,300,532	3,858,991	4,433,234

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 38: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	615,045	487,737	547,894	800,412	1,244,001	2,666,049	4,280,122	6,085,071	8,079,703	10,262,775
Inventory	133,333	160,000	186,667	213,333	240,000	266,667	293,333	320,000	346,667	373,333
Accounts receivable	333,333	400,000	466,667	533,333	600,000	666,667	733,333	800,000	866,667	933,333
Total assets traded	1,081,712	1,047,737	1,201,227	1,547,079	2,084,001	3,599,383	5,306,789	7,205,071	9,293,036	11,569,441
Non-current assets										
Fixed assets (net)	3,712,306	3,393,134	3,068,080	2,737,026	2,399,850	2,056,426	1,706,630	1,350,330	987,396	617,692
Total non-current assets	3,712,306	3,393,134	3,068,080	2,737,026	2,399,850	2,056,426	1,706,630	1,350,330	987,396	617,692
Total assets	4,794,018	4,440,870	4,269,307	4,284,105	4,483,850	5,655,809	7,013,419	8,555,402	10,280,432	12,187,133
Liabilities										
Accounts Payables	367,800	431,216	494,739	558,374	622,124	685,996	749,994	814,123	878,389	942,796
Remaining amount of Loan	549,032	601,081	658,063	720,447	0-	0-	0-	0-	0-	0
Total current liabilities	916,832	1,032,296	1,152,802	1,278,821	622,124	685,996	749,994	814,123	878,389	942,796
Non-current liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
long term loans	1,979,591	1,378,510	720,447	0	0	0	0	0	0	0
Total long - term liabilities	1,979,591	1,378,510	720,447	0	0	0	0	0	0	0
Total liabilities	2,896,423	2,410,807	1,873,249	1,278,821	622,124	685,996	749,994	814,123	878,389	942,796
Property rights										
Members Contributions	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076	2,020,076
Compulsory reserve	0	13,247	49,846	110,769	196,413	307,222	436,583	584,368	750,445	934,674
Profit (loss)	122,482-	3,260-	326,135	874,438	1,645,237	2,642,515	3,806,765	5,136,834	6,631,522	8,289,586
Total equity	1,897,595	2,030,064	2,396,058	3,005,284	3,861,726	4,969,813	6,263,424	7,741,278	9,402,043	11,244,337
Total liabilities and equity	4,794,018	4,440,870	4,269,307	4,284,105	4,483,850	5,655,809	7,013,419	8,555,402	10,280,432	12,187,133

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 39: Free Net Cash flows Table

Net Cashflows	Pre-Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-1,836,118	392,262	476,701	485,947	495,198	504,429	513,612	528,154	543,099	558,459	574,243	8,843,515
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash Flows	-1,836,118	288,741	316,827	291,615	268,314	246,779	226,875	210,648	195,578	181,583	168,587	2,596,290

Table below (40) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 40: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free											
Cashflows	-1,836,118	392,262	476,701	485,947	495,198	504,429	513,612	528,154	543,099	558,459	574,243
Project Value	1,836,118	1,443,856	967,155	481,208	-13,990	-518,419	-1,032,031	-1,560,184	2,103,284	2,661,74	3,235,986
Payback Period	3	1	1	1	0	0	0	0	0	0	0

Table 41: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	3,155,719
Payback Period	3
Internal Rate of Return	%31.54

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,155,719 Jordan Dinars considering that the project provides 10 Job Opportunities for the governorate residents.

Table 42: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%31.54	3	10.75%
Revenues declined by 10%	%30.73	4	10.75%
Operating Expenses Increased by 10%	%19.28	7	10.75%